R Studio != R

(**!=** je R sintaksa za “ni enako” ☺)

Orodje za uporabo R-a

(**IDE** = Integrated Development Environment)

Programski jezik

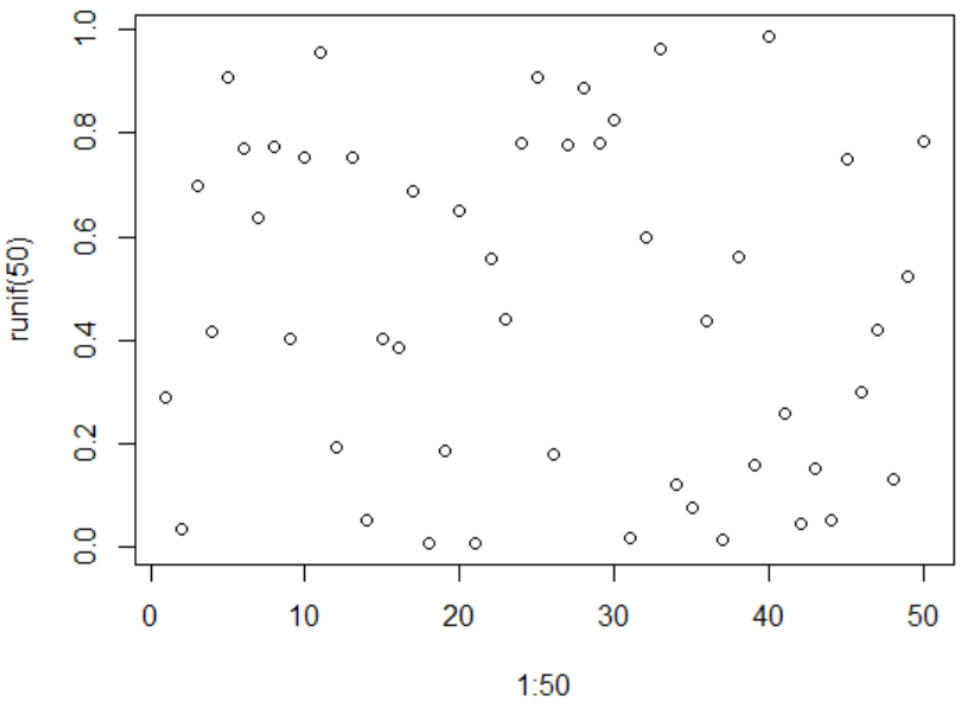
RGui (64-bit)

File History Resize Windows

R Console

```
R version 3.6.3 (2020-02-29) -- "Holding the Windsock"
Copyright (C) 2020 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Previously saved workspace restored]

> plot(x=1:50, y=runif(50))
>
```

R Graphics: Device 2 (ACTIVE)



WolfMonitoringSLO1920 - RStudio

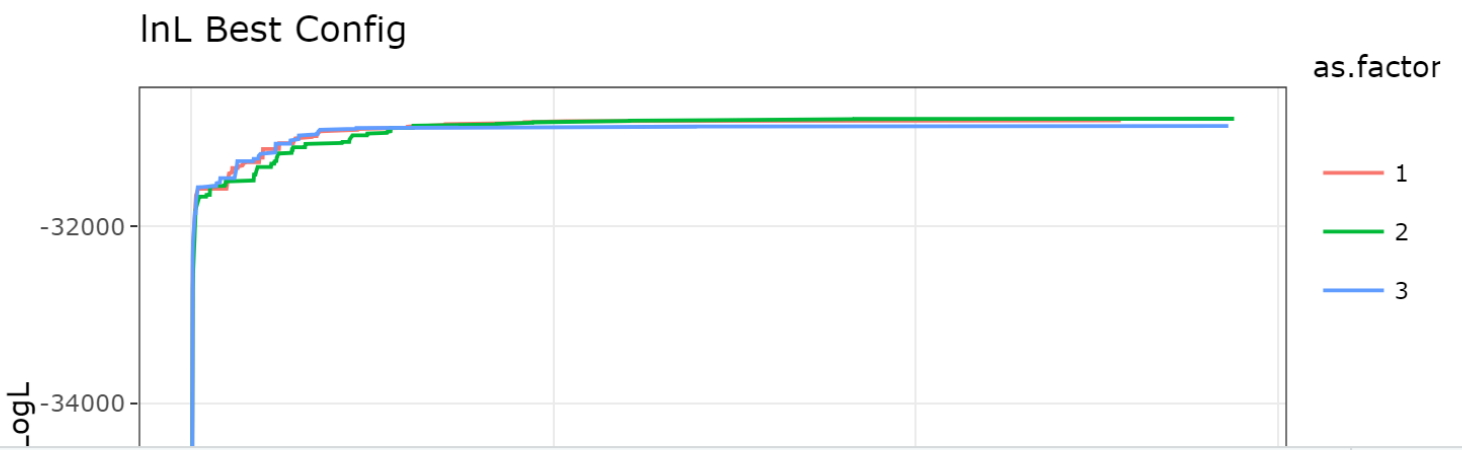
File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | WolfMonitoringSLO1920

s.Rmd | WSlo1920_01_Hybridization.Rmd | WSlo1920_02_ColonyDataPrep.Rmd | WSlo1920_04_Parentage.Rmd | coordconvert

Preview | Insert | Run

```
51 ```
52
53 Graph the PLS run.
54 ```{r}
55 #library(plotly)
56 pp=ggplot(aes(x=NumIterate,color=as.factor(Run)),data=midresults_poly)+theme_bw()
57 plp=ggplotly(pp+geom_line(aes(y=BtLogL))+ggtitle("lnL Best Config"))
58 plp
59 ```
```



742:1 | PC 45 - Menina | R Markdown

Console | Terminal | Jobs

~/R/WolfMonitoringSLO/WolfMonitoringSLO1920/

```
help.start() for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/R/WolfMonitoringSLO/WolfMonitoringSLO1920/.RData]

>
```

Environment | History | Connections | Tutorial

Import Dataset | List

Global Environment

Data

bestconfig	370 obs. of 1 variable
fadTable_P	394 obs. of 5 variables
fullsibs	1021 obs. of 3 variables
kinshipTab…	394 obs. of 8 variables
maternity	301 obs. of 7 variables
midresults…	8375 obs. of 27 variables
parents	120 obs. of 1 variable
parents_da…	120 obs. of 39 variables
paternity	289 obs. of 7 variables
ped.organi…	List of 2
ped.organi…	List of 2
plp	List of 8

Files | Plots | Packages | Help | Viewer

New Folder | Delete | Rename | More

Home > R > WolfMonitoringSLO > WolfMonitoringSLO19

Name	Size	Modifie
Structure		
test.png	56.9 KB	Nov 3
Test_Kindepth.R	1.1 KB	Nov 3
TissueSamples2020.xlsx	55.7 KB	Nov 2
WolfDinIt_Dog_Hybridization.Rmd	673 B	Oct 2
WolfMonitoringSLO1920.Rproj	218 B	Dec 1
WSlo1920_00_Main.nb.html	657 KB	Oct 3

WolfMonitoringSLO1920 - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | WolfMonitoringSLO1920

.Rmd | WSlo1920_01_Hybridization.Rmd | WSlo1920_02_ColonyDataPrep.Rmd | WSlo1920_04_Parentage.Rmd | coordconvert

Preview | Insert | Run

```
51 ```
52
53 Graph the PLS run.
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57 plp=ggplotly(pp+geom_line(aes(y=BtLogL))+ggtitle("lnL Best Config"))
58 plp
59 ```
```
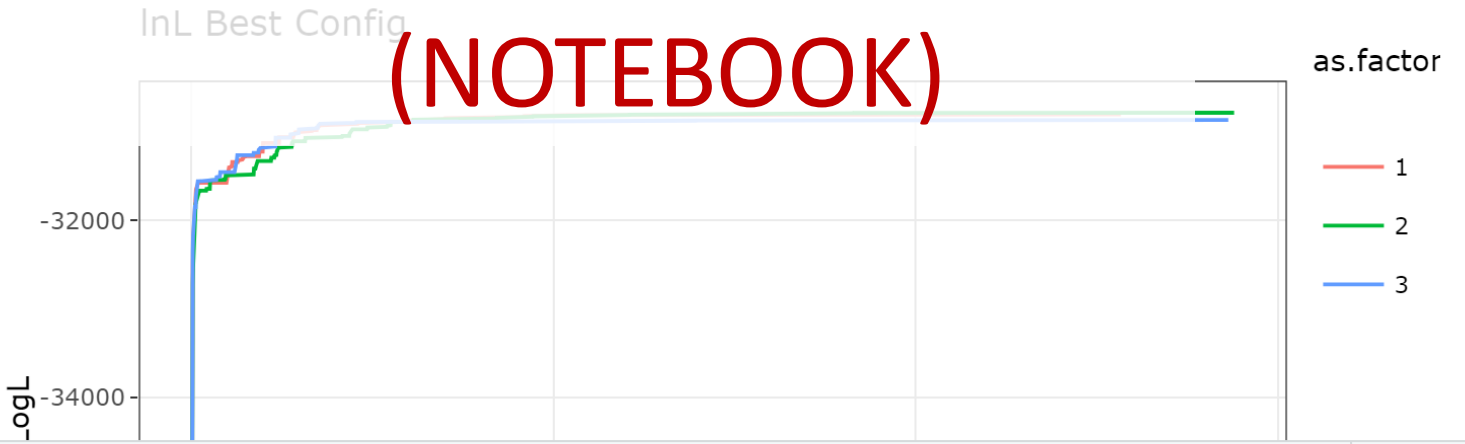

UREJEVALNIK BESEDILA (NOTEBOOK)



742:1 | PC 45 - Menina | R Markdown

Console | Terminal | Jobs

~/R/WolfMonitoringSLO/WolfMonitoringSLO1920/

```
Type 'q()' to quit R.

[Workspace loaded from ~/R/WolfMonitoringSLO/WolfMonitoringSLO1920/.RData]

>
```

KONZOLA (DEJANSKI R!)

Environment | History | Connections | Tutorial

Import Dataset | List

Global Environment

Data

bestconfig	370 obs. of 1 variable
fadTable_P	394 obs. of 5 variables
fullsibs	1021 obs. of 3 variables
kinship	394 obs. of 5 variables
maternity	1021 obs. of 7 variables
midresults…	8375 obs. of 27 variables
parents	110 obs. of 1 variable
parents_da…	110 obs. of 9 variables
paternity	289 obs. of 7 variables
ped.organi…	List of 2
ped.organi…	List of 2
plp	List of 8

OKOLJE (objekti)

Files | Plots | Packages | Help | Viewer

New Folder | Delete | Rename | More

Home > R > WolfMonitoringSLO > WolfMonitoringSLO19

Name	Size	Modifie
Structure		
test.p	56.0 KB	Nov 3
Test_Kinde	KB	Nov 3
TissueSamples2020.xlsx	55.7 KB	Nov 2
Wolf	673 B	Oct 2
Wolf	318 B	Dec 1
WSlo1920_00_Main.nb.html	657 KB	Oct 3

DATOTEKE POMOČ...

UREJEVALNIK – R Notebook

ConsGen2016.PopStructure.Rmd

Knit HTML Run

```
138  X=scaleGen(cheetah,NA.method="mean")
139  pca1 <- dudi.pca(X,cent=FALSE,scale=FALSE, scannf=FALSE, nf=2) #make PCA
140  ```
141
142  Take a look at the eigenvalues and plot the data.
143
144  ```{r}
145  #plot eigenvalues
146  par(mfrow=c(1,1))
147  barplot(pca1$eig[1:50],main="PCA Eigenvalues", col=heat.colors(50))
148
149  #let's look a the data
150  s.label(pca1$li)
151
152  #hmmm... some clustering. Let's mark countries.
153  s.class(pca1$li, other(cheetah)$Origin)
154
155  #Do these conform to geographic regions?
156  s.class(pca1$li, other(cheetah)$Area)
157
158  #Each geographic region seems to be a different populaton - let's write this into
     our object with genetic data.
159  pop(cheetah)=other(cheetah)$Area
160
161  ```
162
163  PCA nicely shows genetic structure.
164  Let's calculate Wright's Fst.
165
166  ```{r cheetahFST, warning=F, message=F}
167  #install.packages("hierfstat")
168  library("hierfstat")
```

142:50 EXERCISE: STRUCTURE OF CHEETAH POPOULATIONS R Markdown

Zapiski, misli, interpretacija – karkoli želite.

„Chunk“-i kode – to se požene in naredi rezultat.

```{r ime_chunka}
....koda
....koda
```  #konec chunka

ClimateChange_Priprava - RStudio

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Vaja5_KlimatskeSpremembe.Rmd

Preview | Insert | Run

```
_____________________________
### Naloga 2.1

Nariši graf rasti človeške populacije v Holocenu. Potem pa še en graf za zadnjih 200 let. Graf za zadnjih 200 let "spimpaj", da bo lep in shrani za kasneje kot `graph_pop`.

```{r fig.width=4, fig.height=3}

PopGrowth %>% ggplot(aes(x=Year, y=WorldPopulation)) +
  geom_line()+xlim(-10000, 2020)

```


Ali nas je kaj več? :)

____
Naloga 2.2

Ob tem, da človeštvo raste, pa se tudi razvijamo in vedno več trošimo. Poglejmo še to.

- Naloži podatke iz `CO2Emissions_World.csv` v objekt `co2world`. Naredi graf rasti skupnih CO2 izpustov skozi čas in mu daj ime `graph_co2`, pa še en graf za izpuste CO2 na prebivalca in mu day ime `graph_co2_percapita`. Oba grafa izriši.

Podatki so pripravljeni iz podatkov "Our World in Data", več o podatkih kasneje

Postopek (dogovor):
```

‘Chunk’ kode, ki lahko nekaj naredi.

220:1 | Chunk 8 | R Markdown

Console

ClimateChange_Priprava - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | ClimateChange_Priprava

Vaja5_KlimatskeSpremembe.Rmd

Preview | Insert | Run

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### Naloga 2.1

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```


Ali nas je kaj več? :)

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Ob tem, da človeštvo raste, pa se tudi razvijamo in vedno več trošimo. Poglejmo še to.

- Naloži podatke iz `CO2Emissions_World.csv` v objekt `co2world`. Naredi graf rasti skupnih CO2 izpustov skozi čas in mu daj ime `graph_co2`, pa še en graf za izpuste CO2 na prebivalca in mu day ime `graph_co2_percapita`. Oba grafa izriši.

Podatki so pripravljeni iz podatkov "Our World in Data", več o podatkih kasneje

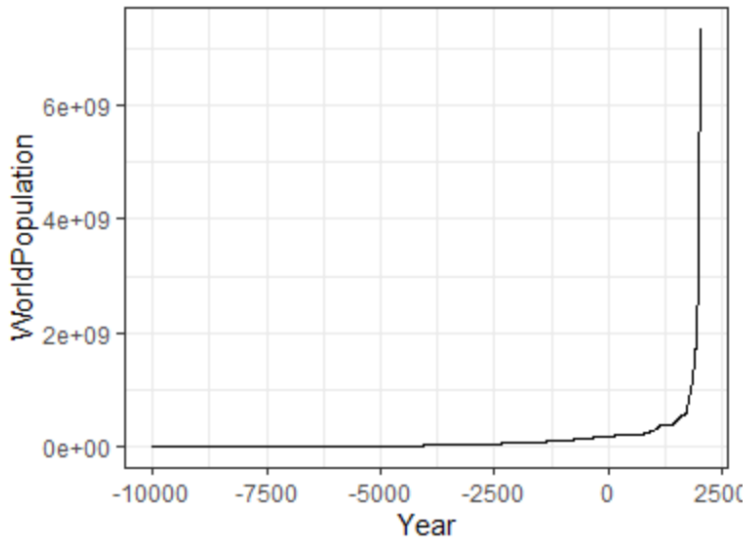
Postopek (dogovor):
```

<CTRL – ENTER> (vrstica)

<CTRL – SHIFT – ENTER> (chunk)

Pošlje R procesorju v konzolo!

220:1 | Chunk 8 | R Markdown

Console

ClimateChange_Priprava - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | ClimateChange_Priprava

Vaja5_KlimatskeSpremembe.Rmd*

Preview | Insert | Run

```
### Naloga 2.1

Nariši graf rasti človeške populacije v Holocenu. Potem pa še en graf za zadnjih 200 let. Graf za zadnjih 200 let "spimpaj", da bo lep in shrani za kasneje kot `graph_pop`.

```{r fig.width=4, fig.height=3}

PopGrowth %>% ggplot(aes(x=Year, y=WorldPopulation)) +
  geom_line()+xlim(-10000, 2020)

```
```



R procesor vrne rezultat.

DOKLER KODE NE ZAŽENETE,
SE NE ZGODI NIČ!!

224:1 Chunk 8 R Markdown

Console

Vektorji – enodimenzionalne podatkovne strukture

```
> a=c(1, 2.5, 4, 8.2, 8, 332, 66, 88, 0.2, -5)
```

1	**2.5**	**4**	**8.2**	**8**	**332**	**66**	**88**	**0.2**	**-5**	Vektor a
1	2	3	4	5	6	7	8	9	10	Pozicija (indeks)

Vektorji

```
> a=c(1, 2.5, 4, 8.2, 8, 332, 66, 88, 0.2, -5)
```

1	2.5	4	8.2	8	332	66	88	0.2	-5	Vektor a
1	2	3	4	5	6	7	8	9	10	Pozicija (indeks)

```
> a[4]
[1] 8.2
```

8.2
1

Vektorji

```
> a=c(1, 2.5, 4, 8.2, 8, 332, 66, 88, 0.2, -5)
```

1	2.5	4	8.2	8	332	66	88	0.2	-5	Vektor a
1	2	3	4	5	6	7	8	9	10	Pozicija (indeks)

```
> a[c(6,7,8,9)]
[1] 332 66 88 0.2
```

332	66	88	0.2
1	2	3	4

Vektorji

```
> a = c("muc", "pes", "polž")
```

muc	pes	polž
1	2	3

Vektor a

Pozicija (indeks)

```
> a[3]
[1] "polž"
```

polž
1

Vektorji

Enodimenzionalne podatkovne strukture.

```
> a = c(1,2,3)
> a*2
[1] 2 4 6
```

Naredili smo vektor ‚a‘ in ga množili z 2, dobili nov vektor.

```
> a = c("muc", "pes", "polž")
> paste(a, "ek", sep="")
[1] "mucek"  "pesek"  "polžek„
```

Besedilo gre VEDNO V NAREKOVAJIH!

```
> a = c(1,2,3)
> a == 2
[1] FALSE  TRUE FALSE
```

Rezultat logičnega operatorja je logični vektor.

Vector -> Dataframe

```
> vrsta = c("muc", "pes", "polž")
> stevilo = c(10, 12, 8)
> ima_bolhe = c(TRUE, TRUE, FALSE)
```

Vector -> Dataframe

```
> vrsta = c("muc", "pes", "polž")
> stevilo = c(10, 12, 8)
> ima_bolhe = c(TRUE, TRUE, FALSE)
```

Vektorji

vrsta	muc	pes	polž
stevilo	10	12	8
ima_bolhe	TRUE	TRUE	FALSE

Vector -> Dataframe

```
> vrsta = c("muc", "pes", "polž")
> stevilo = c(10, 12, 8)
> ima_bolhe = c(TRUE, TRUE, FALSE)
```

Vektorji

vrsta

stevilo

ima_bolhe

muc	pes	polž
10	12	8
TRUE	TRUE	FALSE

```
> bolhe = data.frame(vrsta,stevilo,ima_bolhe)
```

Dataframe

bolhe

vrsta	stevilo	ima_bolhe
muc	10	TRUE
pes	12	TRUE
polž	8	FALSE

Dataframe

bolhe

vrsta	stevilo	ima_bolhe
muc	10	TRUE
pes	12	TRUE
polž	8	FALSE

← names

← data

Dataframe

bolhe

vrsta	stevilo	ima_bolhe
muc	10	TRUE
pes	12	TRUE
polž	8	FALSE

← names

← data

```
> names(bolhe)
[1] "vrsta"     "stevilo"   "ima_bolhe"
```

Vektor imen stolpcev…

Dataframe

bolhe

vrsta	stevilo	ima_bolhe
muc	10	TRUE
pes	12	TRUE
polž	8	FALSE

← names

← data

```
> names(bolhe)
[1] "vrsta"     "stevilo"   "ima_bolhe"
```

Vektor imen stolpcev…

```
> names(bolhe) = c("species","num","has_fleas")
```

… ki jih lahko zamenjamo (preimenujemo stolpce).

Dataframe

bolhe

species	num	has_fleas
muc	10	TRUE
pes	12	TRUE
polž	8	FALSE

Indeks vrstic

Indeks stolpcev

Dataframe: bolhe

	1	2	3
	vrsta	stevilo	ima_bolhe
1	muc	10	TRUE
2	pes	12	TRUE
3	polž	8	FALSE

ime_dataframea[**vrstice**, **stolpci**]

Dobimo posamezne podatke

Indeks vrstic

Indeks stolpcev

Dataframe: bolhe

	1	2	3
	vrsta	stevilo	ima_bolhe
1	muc	10	TRUE
2	pes	12	TRUE
3	polž	8	FALSE

```
>bolhe[1, 1]
[1] "muc"
```

„Iz dataframe-a vzami prvi stolpec, prvo vrstico“

Indeks vrstic Indeks stolpcev

Dataframe: bolhe

	1	2	3
	vrsta	stevilo	ima_bolhe
1	muc	10	TRUE
2	pes	12	TRUE
3	polž	8	FALSE

```
>bolhe[1,]
   vrsta stevilo ima_bolhe
1 "muc"      2      TRUE
```

„Iz dataframe-a vzami prvo vrstico, vse stolpce“

Indeks vrstic

Indeks stolpcev

Dataframe: bolhe

	1	2	3
	vrsta	stevilo	ima_bolhe
1	muc	10	TRUE
2	pes	12	TRUE
3	polž	8	FALSE

```
> bolhe[,1]
[1] "muc"  "pes"  "polž"
```

„Iz dataframe-a vzami prvi stolpec

Indeks vrstic

Indeks stolpcev

Dataframe: bolhe

	1	2	3
	vrsta	stevilo	ima_bolhe
1	muc	10	TRUE
2	pes	12	TRUE
3	polž	8	FALSE

```
> bolhe$vrsta
[1] "muc"  "pes"  "polž"
```

„Iz dataframe-a vzami stolpec ‚vrsta‘“

Indeks vrstic

Indeks stolpcev

Dataframe: bolhe

	1	2	3
	vrsta	stevilo	ima_bolhe
1	muc	10	TRUE
2	pes	12	TRUE
3	polž	8	FALSE

```
> bolhe[ , “vrsta“]
[1] "muc"  "pes"  "polž"
```

„Iz dataframe-a vzami stolpec ‚vrsta‘ (malo drugače)“

Indeks vrstic Indeks stolpcev

Dataframe: bolhe

	1	2	3
	vrsta	stevilo	ima_bolhe
1	muc	10	TRUE
2	pes	12	TRUE
3	polž	8	FALSE

```
> bolhe[ , c("vrsta","stevilo")]
  vrsta stevilo
1   muc       2
2   pes      12
3  polž       4
```

„Iz dataframe-a vzami vse vrstice, stolpca ‚vrsta‘ in ‚stevilo‘“

Indeks vrstic Indeks stolpcev

Dataframe: bolhe

	1	2	3
	vrsta	stevilo	ima_bolhe
1	muc	10	TRUE
2	pes	12	TRUE
3	polž	8	FALSE

```
> bolhe[ , c(1, 2)]
  vrsta stevilo
1   muc       2
2   pes      12
3   polž      4
```

„Iz dataframe-a vzami vse vrstice, stolpca 1 in 2“